Azwad Tamir

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Career Summary

- I have completed my PhD in Electrical Engineering at the Department of ECE at the University of Central Florida in **December 2024** with a concentration in AL/ML.
- My research areas include developing Machine learning architecture for low-data applications in **CV, NLP, 3D IC design**, etc.
- While employed as a Research Assistant at UCF, I have worked on numerous research projects involving the implementation and development of Machine Learning algorithms on various applications including 3D IC Design, Bioinformatics, Computer Vision, NLP, and Drug Discovery.
- I have multiple publications in **top ML journals and conferences** and have experience working in a research environment.
- I have also worked as a Teaching Assistant at the department of ECE for 3+ years and have served as the instructor for Lab courses along with hosting several guest lectures on multiple courses.
- I also have experience in drafting project proposals for joint research projects with various companies including **Tokyo Electron, ARM, and Synopsys**.

Education

University of Central Florida (UCF) *Orlando, Florida, USA*

PhD in Electrical Engineering *Fall 2019 - Fall 2024*

- **Research Areas:** Developing Transformer based ML Models for low data Applications
- **Dissertation Title:** Architectural and Training regime Modifications to Transformer models for Low Data Applications
- **Current CGPA:** 4.0 out of 4.0
- **Courses:** Machine Learning, Machine Learning Methods for Biomedical Data, Computational Genomics, Data Mining Methodology II, Intro to Neural Networks, Computer Vision, 3D Computer Vision, Design & Analysis of Algorithms, CMOS Analog & Digital Circuit Design, Malware Software Vulnerability, Incident Response Technologies

University of Central Florida (UCF) *Florida, USA*

Masters of Science in Electrical Engineering *Aug 2019 - Spring 2023*

- **CGPA:** 4.00 out of 4.00
- **Major:** Electical Engineering

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*

BSc in Electrical and Electronic Engineering *Feb 2013 - Aug 2017*

- **CGPA:** 3.54 out of 4.00
- **Major:** Electronics

Experience

University of Central Florida (UCF) *Orlando, USA*

Teaching Assistant *Aug 2021 - Dec 2024*

Primary Responsibilities include:

- Helping the course instructor in preparing course contents and syllabus.
- Preparing course materials like lab manuals and assignments.
- Conducting Laboratory courses.
- Grading assignments and exam papers.
- Taking problem solving sessions and guest lectures.

Bayer Research and Development Services *Missouri, USA*

Remote Sensing Data Science Intern *May 2023 - Aug 2023*

Project:

- Determining Phenotype of corn plants from UAV (Unmanned Ariel Vehicle) images of corn fields.
- Implemented various CNN and transformer-based vision algorithms and evaluated and compared their performance to identify the best-suited model for the application.

University of Central Florida (UCF) *Florida, USA*

Research Assistant *Aug 2019 - Present*

Primary Responsibility includes working on various ML related research Projects. Some of them are:

- **KITE_DDI:** A Knowledge graph integrated Transformer architecture for accurately predicting Drug-Drug Interaction events from SMILES
- **Prot_EC:** A Transformer based deep learning model to accurately predict Enzyme Commission numbers from full-scale protein sequences
- Analytical Quadratic, Reinforcement Learning and **GPT-GNN** based Machine learning graph partitioners for **3D IC design**.
- Collaborative Project with **Tokyo Electron Ltd**. to develop Design Flow for 3D IC design

Southeast University *Dhaka, Bangladesh*

Lecturer, Department of Electrical and Electronic Engineering *March 2018 - Aug 2019*

- Conducted courses and Labs related to C/C++ and assembly programming languages.
- Acted as the coordinator for the EEE department.
- Served in the committee for Enhancing Educational Experience through OBE.
- Moderated the SEU club for Engineering Innovations.

IEEE BUET Student Branch *Dhaka, Bangladesh*

Treasurer, IAS chapter *Aug 2015 - Aug 2017*

- Acted as the university liaison with the Industry and Organized study trips to Engineering Facilities and Factories of different Companies.
- Served in the organizing committees of conferences hosted by IEEE BUET

Selected Research Projects

KITE-DDI *UCF, USA*

A Knowledge graph integrated Transformer model for accurately predicting Drug-Drug Interaction Events

- A Knowledge Graph and transformer based fusion model have been proposed which could predict DDI events with SOTA accuracy.
- Shows robust performance with little dependency on sequence length and training set size.
- Robust model requiring very few hyperparameter optimization and takes raw Drug SMILES as input.
- GitHub Repository: `https://github.com/azwad-tamir/KiteDDI`

ProtEC *UCF, USA*

A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers

- A Bert-based Transformer model to predict all four Enzyme Commission numbers from full-scale Protein sequences.
- Pretrained ProtBert modules were selectively finetuned on the Uniprot SwissProt protein database to achieve SOTA accuracy.
- GitHub Repository: `https://github.com/azwad-tamir/Prot_EC`
- Paper Link: `https://ieeexplore.ieee.org/document/10238761`

ProtGO *UCF, USA*

A Transformer Based Fusion model for accurately predicting Gene Ontology (GO) Terms from full-scale protein sequences.

- A Bert-based Transformer model to annotate full scale protein sequences with Gene Ontology (GO) terms using a classification type parallel ML fusion architecture.
- The proposed model is able to predict GO terms with better accuracy compared to other state-of-the-art methods and could understand motifs and protein 2D structures.
- GitHub Repository: `https://github.com/azwad-tamir/ProtGO`

Machine Leaning based partitioner for 3D IC Design *UCF, USA*

Generative Graph algorithms

- A Generative Pre-Training Graph Neural Network (GPT-GNN) model was used to develop a graph partitioner for 3D-IC design.
- The Gate-level Verilog netlist of the input digital circuit was converted into a graph and fed into GPT-GNN to compute the latent node vectors.
- K-means algorithm was applied on the latent vectors to divide the graph into multiple clusters.
- GitHub Repository: `https://github.com/azwad-tamir/gpt_gnn_3D_partitioner`

Skin Cancer Detection *UCF, USA*

Medical Imaging using ML

- Several Deep Learning classification Model was implemented to detect skin cancer from images of skin lesions.
- Transfer Learning was used on different Vision Transformer (ViT) models to classify skin cancer types on the HAM10000 dataset.
- ViT-based models showed significant improvement over the CNN models.
- GitHub Repository: `https://github.com/azwad-tamir/Transfer_ViT`

ML Forecasting *UCF, USA*

Crime Prediction and Forecasting using Machine Learning Algorithms

- Several Machine Learning models have been designed and implemented to predict future crime.
- The data has been preprocessed and curated to create a clean dataset for training the ML models.
- The results show state of the art performance with the model's ability to predict and forecast future crimes with high accuracy.
- Paper link: `https://www.researchgate.net/publication/355872171_Crime_Prediction_and_Forecasting_using_Machine_Learning_Algorithms`

Street House Number Detection System *UCF, USA*

Computer Vision

- A Machine Learning model was developed that could detect house numbers from street view raw images from the SVHN dataset.
- The model consisted of a localizer based on Yolo-V2 that could detect the areas of interest from the raw images.
- The areas of interest are then fed into a CNN-based model to detect the digits in the house number.
- GitHub Repository: `https://github.com/azwad-tamir/Deep_SVHN`

Anemia Detection from Images of the Eye *BUET, Bangladesh*

Biomedical Engineering

- Design of a tool that could detect the presence of Anemia in a patient from images of the Anterior Conjunctiva of the eye using Image Processing.
- The tool was lightweight and could be implemented in a mobile device.
- Paper Link: **Detection of anemia from Image of the anterior conjunctiva of the eye by image processing and thresholding**

Brain-Drive *BUET, Bangladesh*

Biomedical Engineering

- Design of a smart Device that could extract EEG signals from the brain to control Household appliances
- The Device was implemented using a Raspberry Pi Microcontroller and an OLED screen.
- Paper Link: **Brain-drive: A smart driver for controlling digital appliances using cognitive command**

Selected Publications

A. Tamir and J.-S. Yuan, 'ProtGO: A Transformer based Fusion Model for accurately predicting Gene Ontology (GO) Terms from full scale Protein Sequences', arXiv [cs.LG]. 2024. (submitted to IEEE for publication)

A. Tamir and J.-S. Yuan, 'KITE-DDI: A Knowledge graph Integrated Transformer Model for accurately predicting Drug-Drug Interaction Events from Drug SMILES and Biomedical Knowledge Graph', arXiv [cs.LG]. 2024. (submitted to IEEE for publication)

A. Tamir, M. Salem and J. -S. Yuan, "ProtEC: A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers," in IEEE/ACM Transactions on Computational Biology and Bioinformatics, vol. 20, no. 6, pp. 3691-3702, Nov.-Dec. 2023.

A. Tamir, M. Salem, J. Lin, Q. Alasad, and J. Yuan, "Multi-Tier 3D IC Physical Design with Analytical Quadratic Partitioning Algorithm Using 2D P&R Tool," Electronics, vol. 10, no. 16, p. 1930, Aug. 2021, doi:10.3390/electronics10161930.

A. Tamir, E. Watson, B. Willett, Q. Hasan and J. Yuan, "Crime Prediction and Forecasting using Machine Learning Algorithms," International Journal of Computer Science and information Technologies, Vol. 12(2), 2021, pp. 26-33, ISSN:0975-9646.

A. Tamir and M. Saif, "Modelling of Performance Parameters of Single Photon Avalanche Detector Incorporating Dead Space Effects and History Dependent Ionization Coefficient," 2018, Southeast University Journal of Science and Engineering (SEUJSE), ISSN: 1999-1630.

A. Tamir, C. Jahan, M. Saif, S. Zaman, M. Islam, A. Khan, S. Fattah, and C. Shahnaz, "Detection of anemia from image of the anterior conjunctiva of the eye by image processing and thresholding," 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), Dhaka, 2017, pp. 697-701. doi: 10.1109/R10-HTC.2017.8289053.

R. Chowdhury, A. Sun, A. Tamir, C. Shahnaz, and S. Fattah, "Brain-drive: A smart driver for controlling digital appliances using cognitive command," 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), Dhaka, 2017, pp. 851-855. doi: 10.1109/R10-HTC.2017.8289087.

M. Bhuyan and A. Tamir, "Evaluating COs of computer programming course for OBE-based BSc in EEE program," International Journal of Learning and Teaching, vol. 12(2), pp. 86–99, doi:10.18844/ijlt.v12i2.4576.

Skills

Programming	Python(numpy, scipy, pandas, etc.), R, C/C++, Bash, Java, Matlab/Simulink, Verilog, Assembly, Android programming
Frameworks	HuggingFace, PyTorch, TensorFlow, Scikit-learn, Keras
Software	Agile, Synopsys Design & ICC2 Compiler, Lumerical FTDT solutions, Cadence Virtuoso, SPICE, Proteus
Hardware	Microcontroller (AVR), Arduino, Raspberry Pi, FPGA
Language Skills	English, Bengali, German

Teaching Assistant Assignments

Fall 2021 Term:

- **EEL 4768:** Computer Architecture. **Enrollment:** 51 Students
- **EEE 3350:** Semiconductor Devices. **Enrollment:** 65 Students
- **EEE 4314:** Device Electronics for Integrated Circuits. **Enrollment:** 17 Students

Spring 2022 Term:

- **EEL 3290:** Global Energy Issues. **Enrollment:** 33 Students
- **EEE 3350:** Semiconductor Devices. **Enrollment:** 82 Students

Summer 2022 Term:

- **EEE 3342:** Digital Systems Laboratory. **Enrollment:** 19 Students
- **EEE 3350:** Semiconductor Devices. **Enrollment:** 50 Students

Fall 2022 Term:

- **EEE 3307:** Electronics I. **Enrollment:** 51 Students
- **EEE 3342:** Digital Systems Laboratory. **Enrollment:** 24 Students
- **EEE 3123:** Linear Circuits II Laboratory. **Enrollment:** 12 Students

Spring 2023 Term:

- **EEE 3350:** Semiconductor Devices. **Enrollment:** 78 Students
- **EEE 3342:** Digital Systems Laboratory. **Enrollment:** 32 Students

Fall 2023 Term:

- **EEE 3123:** Linear Circuits II Laboratory. **Enrollment:** 65 Students
- **EEE 3307:** Electronics I. **Enrollment:** 46 Students

Spring 2024 Term:

- **EEE 3123:** Linear Circuits II Laboratory. **Enrollment:** 53 Students
- **EEE 3307:** Electronics I. **Enrollment:** 58 Students

Summer 2024 Term:

- **EEE 4309:** Electronics II Laboratory. **Enrollment:** 46 Students
- **EEE 4309:** Electronics II. **Enrollment:** 46 Students

Fall 2024 Term:

- **EEE 4309:** Electronics II Laboratory. **Enrollment:** 26 Students
- **EEE 3350:** Semiconductor Devices. **Enrollment:** 64 Students

Certifications

Stanford University	Machine Learning, course conducted by Coursera
DeepLearning.AI	Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
DeepLearning.AI	Neural Networks and Deep Learning
DeepLearning.AI	Natural Language Processing with Attention Models
DeepLearning.AI	Structuring Machine Learning Projects
UIUC	VLSI CAD Part II: Layout. Course conducted by Coursera

Talks, Peer Reviews & Professional Presentations

Reviewer 2nd International Conference on Emerging Technology and Interdisciplinary Sciences (ICETIS)
Reviewer International Conference on Robotics, Electrical and Signal Processing Techniques hosted by IEEE
Reviewer Southeast University Journal fo Science and Engineering (SEUJSE)
Reviewer Journal of Current Proteomics
Talk Towards Emergence of Low data Machine Learning algorithms for Bioinformatics and Biomedical Applications at the UCF graduate seminar.
Presentation Proposal and Milestone Presentation for the 3D IC joint project between UCF, TEL, ARM and Synopsys.
Presentation Year end exit presentation for the achievements and Deliverables in the joint 3D IC project between UCF and TEL.
Guest Lecture Implementation of 3D IC design using Reinforcement Learning and GPT-GNN, CMOS Analog & digital IC Design Graduate
course at UCF.
Guest Lecture Application of Machine Learning algorithms and Artificial Intelligence in Digital IC Physical Design, Semiconductor Devices
senior level course at UCF.
Guest Lecture Recent Advancements of Machine Learning and Artificial Intelligence in the semiconductor Industry, Electronics I undergraduate course at UCF.
Guest Lecture Understanding the basics of Natural Language Processing (NLP) techniques in recent AI algorithms like ChatGPT, Semiconductor Devices senior level course at UCF.

Achievements

2019	**Fellowship**, Dean's Scholarship	*University of Central Florida*
2019	**321/340 (Q:167, V:154, AWA:4.5)**, Graduate Record Examinations (GRE)	*Bangladesh*
2019	**112/120**, TOEFL iBT	*Bangladesh*
2015	**Bronze Awardee**, Duke of Edinburgh Program	*Bangladesh*
2010	**Outstanding Achievement in GCE O-levels**, The Daily Star Awards	*Dhaka*
2012	**Outstanding Achievement and Distinction in Mathematics in GCE A-levels**, The Daily Star Awards	*Dhaka*

References

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